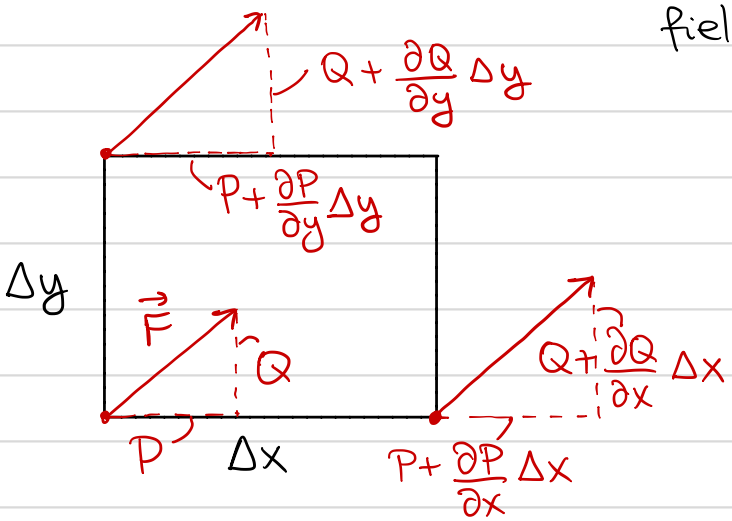
Curl and divergence (16.5)

$\vec{F} = P\hat{i} + Q\hat{j}$ vector field on $\mathbb{R}^2$

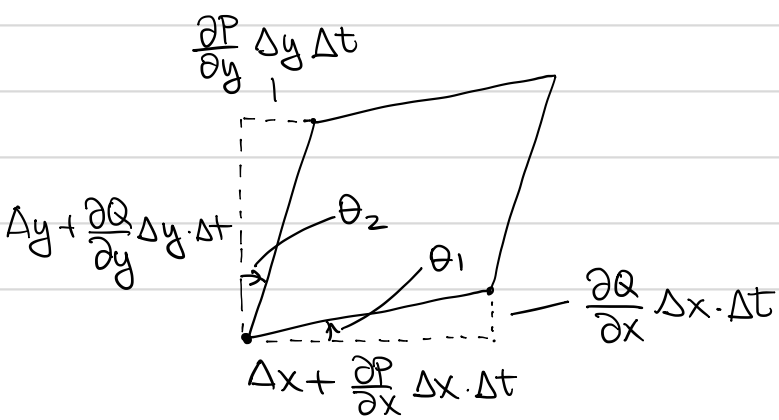
$$\boxed{\begin{array}{l} \operatorname{curl}(\vec{F}) = \dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y} \\ \hline \operatorname{div}(\vec{F}) = \dfrac{\partial P}{\partial x} + \dfrac{\partial Q}{\partial y} \end{array}}$$

Fluid flow interpretation:

Time t: $\vec{F} = (P, Q)$ velocity field



Time $t + \Delta t$:



$$\theta_1 \approx \tan\theta_1 = \frac{\frac{\partial Q}{\partial x}\Delta x \Delta t}{\Delta x + \frac{\partial P}{\partial x}\Delta x \Delta t} \approx \frac{\partial Q}{\partial x}\Delta t$$

$$\theta_2 \approx \frac{\partial P}{\partial y}\Delta t$$

Average angular velocity:

$$\omega = \frac{\theta_1 - \theta_2}{2\,\Delta t} = \frac{1}{2}\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$$

$$= \frac{1}{2}\operatorname{curl}(\vec{F})$$

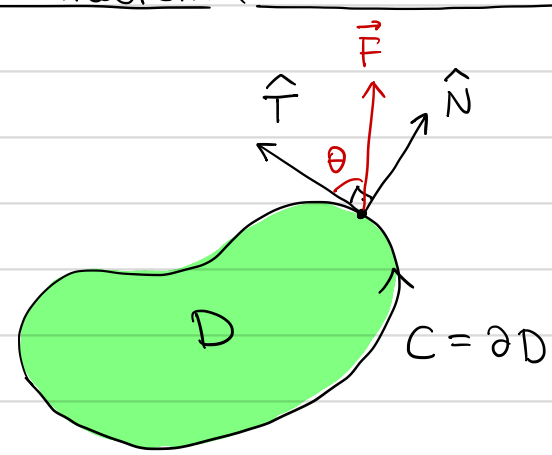
New area:

$$\left(\Delta x + \frac{\partial P}{\partial x}\Delta x \Delta t\right)\left(\Delta y + \frac{\partial Q}{\partial y}\Delta y \Delta t\right)$$

$$\approx \Delta x \Delta y\left(1 + \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right)\Delta t\right)$$

$$= \Delta x \Delta y\,(1 + \operatorname{div}(\vec{F})\,\Delta t)$$

Green's theorem (vector versions)



Parametrization: $\begin{cases} x = x(t) \\ y = y(t) \end{cases}$

$(a \leq t \leq b)$

Unit tangent:

$$\vec{T} = \frac{1}{\sqrt{(x')^2+(y')^2}} \left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle$$

Outward unit normal:

$$\vec{N} = \frac{1}{\sqrt{(x')^2+(y')^2}} \left\langle \frac{dy}{dt}, -\frac{dx}{dt} \right\rangle$$

$\vec{F} = \langle P, Q \rangle$

$ds = \sqrt{(x')^2+(y')^2}\, dt$

$\vec{T}\, ds = \langle dx, dy \rangle = d\vec{r}$

$\vec{N}\, ds = \langle dy, -dx \rangle$

$$\boxed{\underbrace{\int_C \vec{F}\cdot\vec{T}\, ds}_{\text{Circulation of } \vec{F} \text{ around } C} = \iint_D \operatorname{curl}(\vec{F})\, dA}$$

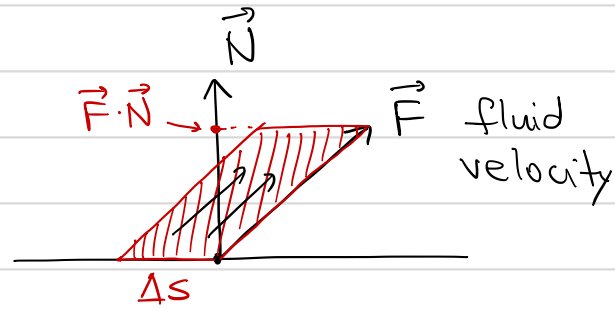
$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

$$\boxed{\underbrace{\int_C \vec{F}\cdot\vec{N}\, ds}_{\text{Flux of } \vec{F} \text{ across } C} = \iint_D \operatorname{div}(\vec{F})\, dA}$$

$$\int_C P\,dy + Q(-dx) =$$

$$= \int_C (-Q)\,dx + P\,dy = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$$

Flux:



$\rho\, \Delta s \cdot (\vec{F}\cdot\vec{N})$ = Mass of fluid out of D through Δs in one second